Strategy Learner - Q-Learner Hints and Tips

Q Trader Hints and Tips

Overview

If you are using a Q-Learner to power your strategy learner then you will leverage the Q-Learner developed earlier in the course. Conceptually, the Strategy Learner project is about writing the "Environment" code that leverages the Q-Learner. As a reminder, a Q-Learner is a Reinforcement Learner, so you will not leverage the Supervised Learners (e.g., DTLearner, RTLearner, BagLearner) developed in earlier assignments.

The indicators define most of the “state” for your learner, an additional component of state you may use is whether or not you are currently holding a position (e.g. long, short, or in cash). The recommended actions at to either move into the desired position `(e.g., Long, Cash, Short)` or to perform a specific trading action `(e.g., Buy, Sell, Do Nothing)`. Some experimentation may be required to determine which type of actions work best for your solution architecture.

Overall, your tasks for this project include

- Build a strategy learner based on your Q-Learner and previously developed indicators.
- Test/debug the strategy learner on specific symbol/time period problems

Implement the Strategy Learner

For this part of the project you should develop a learner that can learn a trading policy using your Q-Learner. You should be able to use your Q-Learner from the earlier project directly, with no changes. You will need to write code in `StrategyLearner.py` to “wrap” your Q-Learner appropriately to frame the trading problem for it. Utilize the template provided in `StrategyLearner.py` Overall the structure of your strategy learner should be arranged as below. Note that this is a suggestion, not a requirement:

For the policy learning part:

- Select several technical features, and compute their values for the training data
- Discretize the values of the features
- Instantiate a Q-learner
- For each day in the training data:
  - Compute the current state (including holding)
  - Compute the reward for the last action
  - Query the learner with the current state and reward to get an action
  - Implement the action the learner returned (LONG, CASH, SHORT), and update portfolio value
- Repeat the above loop multiple times until cumulative return stops improving.

For the policy testing part:

- For each day in the testing data:
  - Compute the current state
  - Query the learner with the current state to get an action
  - Implement the action the learner returned (LONG, CASH, SHORT), and update portfolio value
  - **DO NOT UPDATE Q DURING TESTING**— learning must be turned off in this phase
- Return the resulting trades in a data frame

Training and testing for each situation should run in less than 30 seconds (specifically, the time noted in the assignment instructions). We reserve the right to use different time periods if necessary to reduce auto grading time.